Unit 1: Stress Concentration and Fatigue – Short Notes

These notes cover the important concepts of stress concentration and fatigue in machine components in a short and easy format for quick revision.

1. Stress Concentration

Stress concentration occurs when stress becomes very high at a particular point due to sudden change in shape or size of a component.

Causes: holes, grooves, keyways, sharp corners, cracks, sudden change in cross-section.

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2. Stress Concentration in Different Loads

Tension: Stress increases near holes and notches.
Bending: Maximum stress occurs at outer fibers.
Torsion: Stress concentration appears around grooves and keyways.

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3. Reduction of Stress Concentration

Use fillets, rounded corners, gradual change in section, relief grooves, and better surface finish.

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4. Theoretical Stress Concentration Factor (Kt)

Kt = Maximum Stress / Nominal Stress
It depends on geometry and loading condition.

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5. Notch Sensitivity (q)

It shows how sensitive a material is to notches.
$q = 0$ means no effect of notch.
$q = 1$ means full effect of notch.

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6. Fatigue Stress Concentration Factor (Kf)

$K_f = 1 + q(K_t - 1)$
Used under fluctuating or cyclic loading.

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7. Cyclic Loading

Repeated loading and unloading on a component is called cyclic loading.
Types: completely reversed, repeated, fluctuating loading.

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8. Endurance Limit

Maximum stress a material can withstand for infinite cycles without failure.

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9. S-N Curve

Graph between stress amplitude (S) and number of cycles to failure (N).
Used to study fatigue life.

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10. Factors Affecting Endurance Limit

Loading Factor: Depends on type of load.
Size Factor: Large components have lower endurance strength.
Surface Factor: Better surface finish increases fatigue strength.

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11. Design Consideration for Fatigue

Avoid sharp corners, use smooth finish, proper material selection, and provide safety factor.

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12. Goodman Diagram

Relation between mean stress and alternating stress for safe design.

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13. Modified Goodman Diagram

More practical form of Goodman relation considering ultimate strength.

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14. Soderberg Equation

Conservative design criterion using yield strength.

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15. Gerber Parabola

Parabolic relation between mean and alternating stress.
More accurate but less conservative.

16. Design for Finite Life

Design considering a limited number of load cycles instead of infinite life.

17. Cumulative Fatigue Damage

Damage caused due to varying stress cycles over time.
Palmgren-Miner rule is commonly used.

Quick Revision Table

Topic	Key Point
Stress Concentration	High stress at discontinuities
Kt	Theoretical stress concentration factor
Kf	Fatigue stress concentration factor
S-N Curve	Stress vs cycles graph
Goodman	Fatigue failure criterion
Soderberg	Uses yield strength